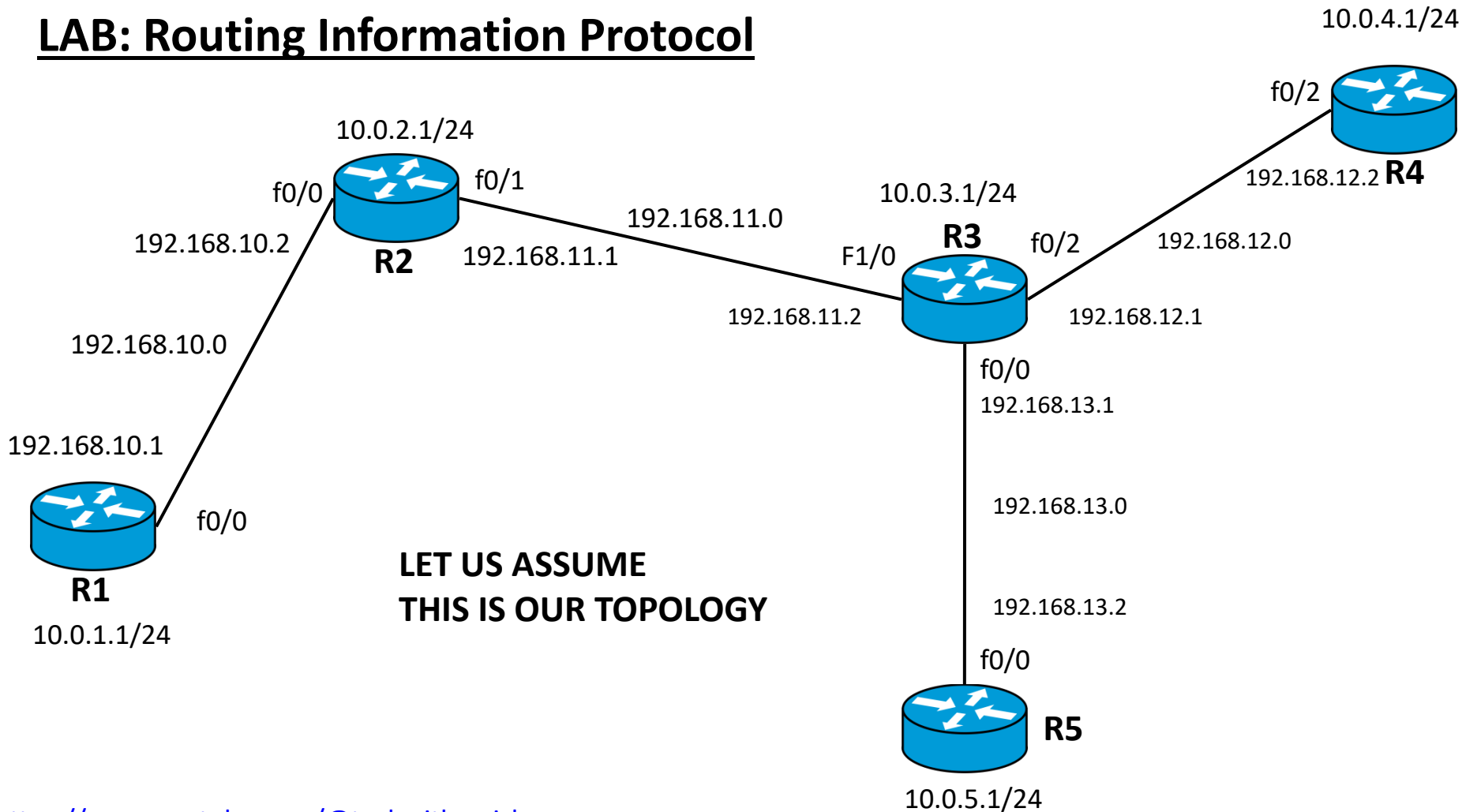


LAB: Routing Information Protocol



STEP 1: In Dynamic Route We have to list all the Directly Connected neighbors:

For R1: 192.168.10.0 and 10.0.1.0

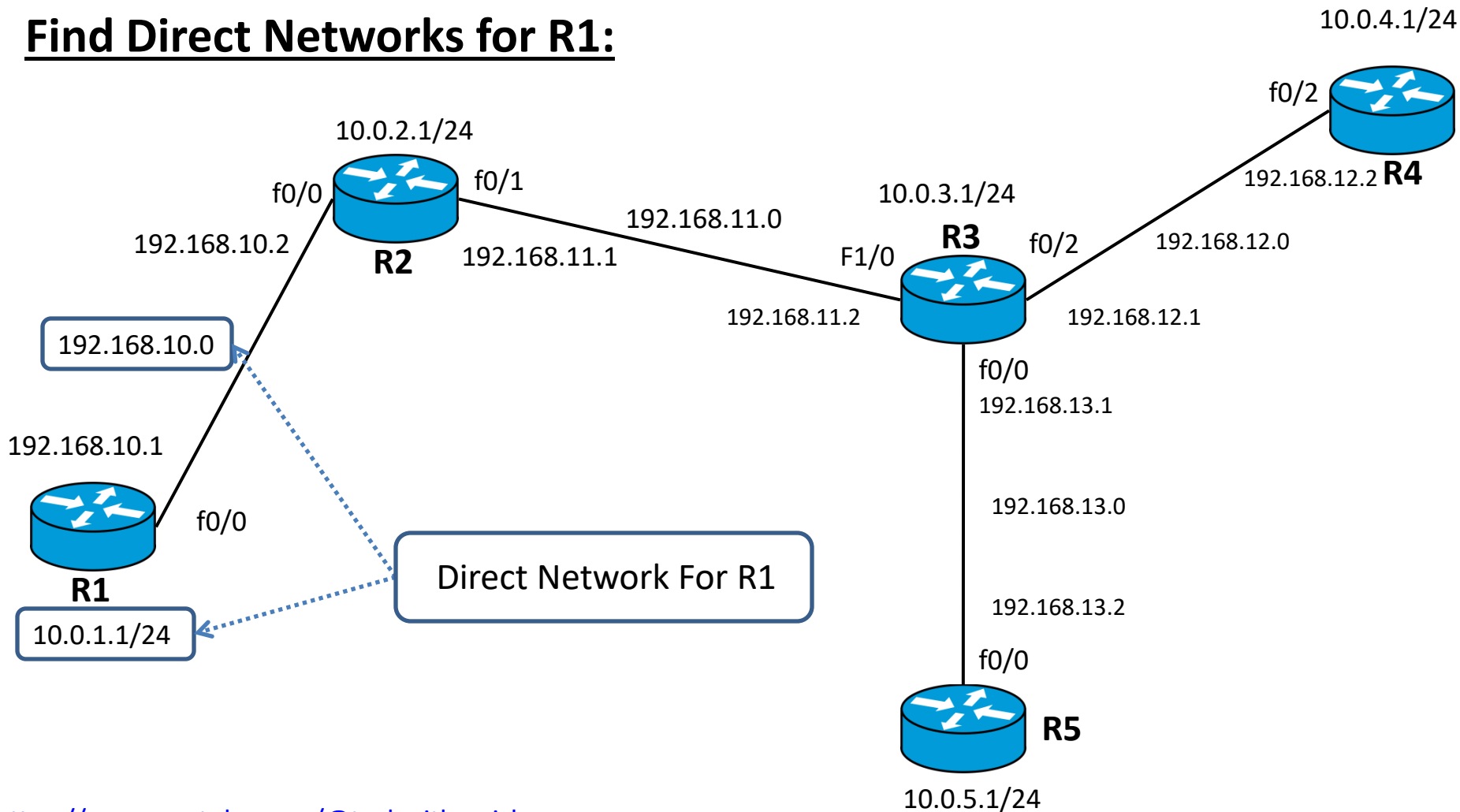
For R2: 192.168.10.0, 192.168.11.0 and 10.0.2.0

For R3: 192.168.11.0, 192.168.12.0 and 192.168.13.0 and 10.0.3.0

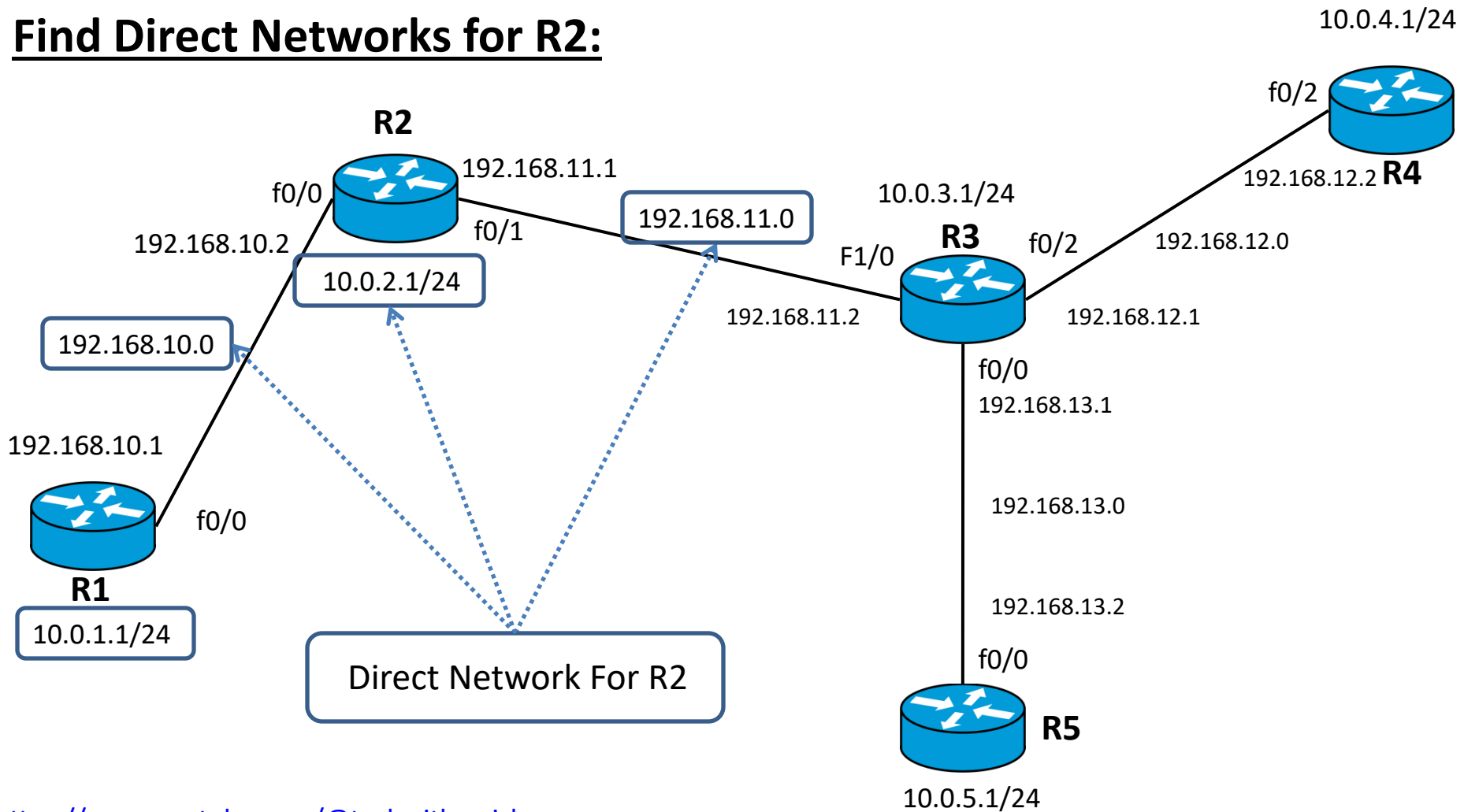
For R4: 192.168.12.0 and 10.0.4.0

For R4: 192.168.13.0 and 10.0.5.0

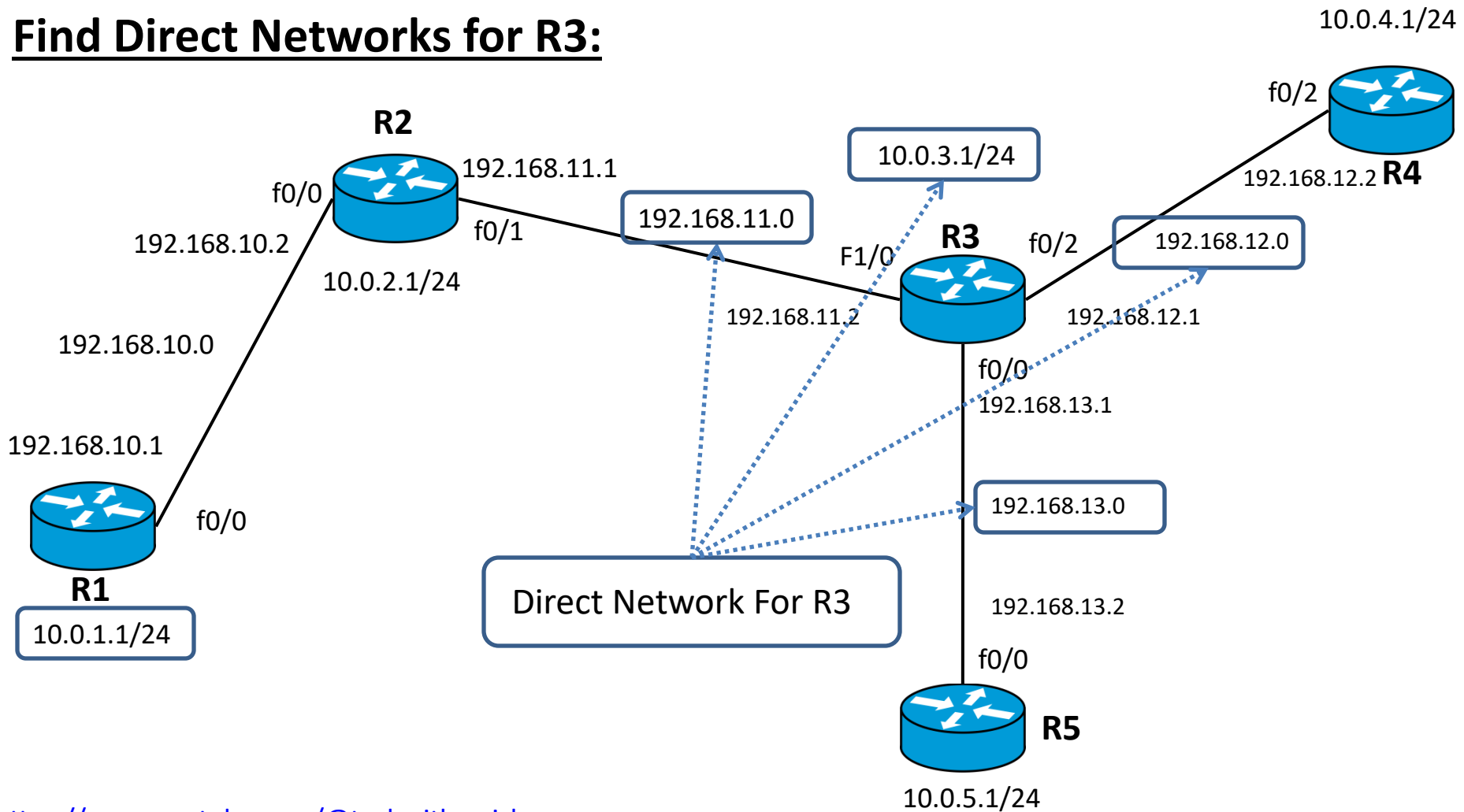
Find Direct Networks for R1:



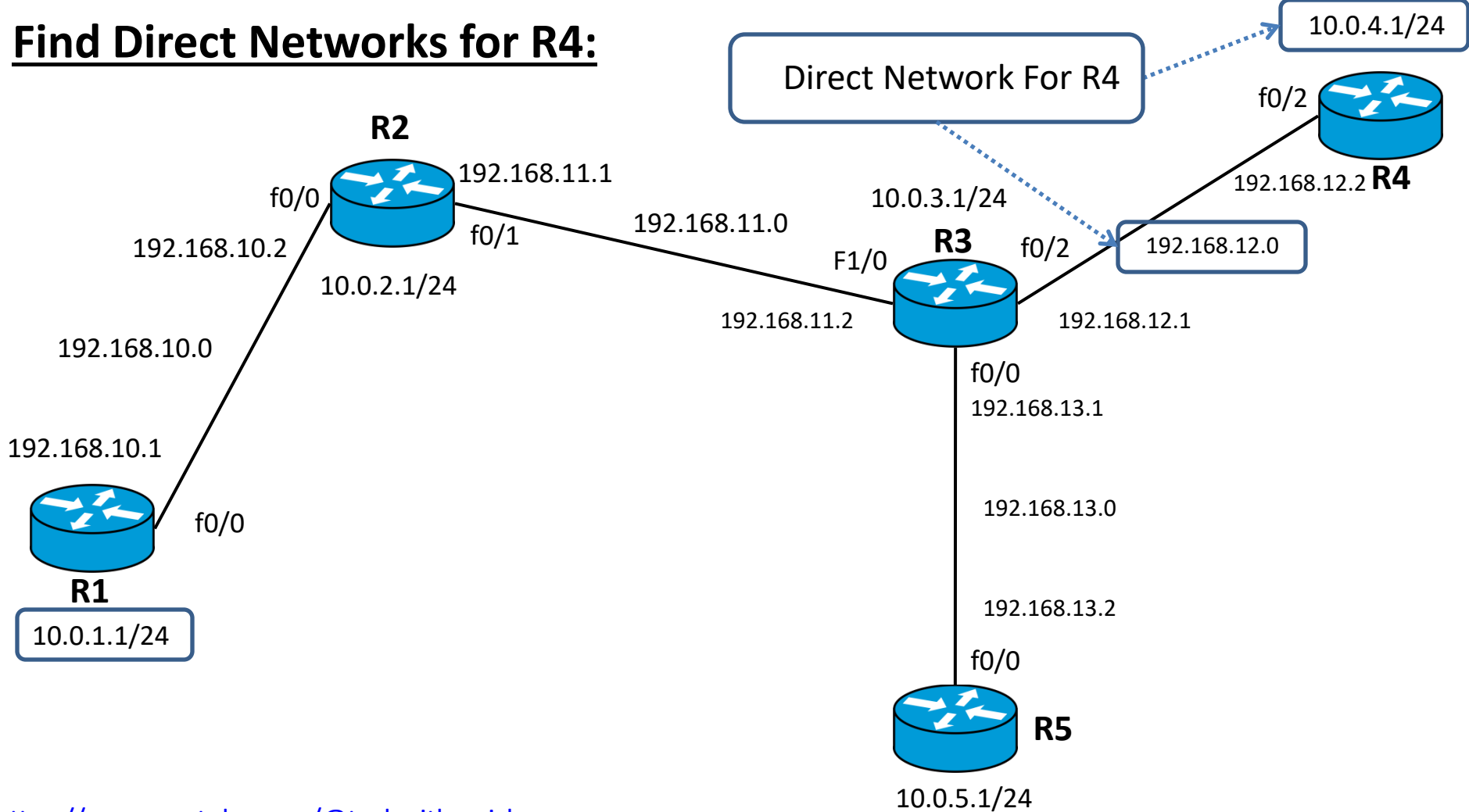
Find Direct Networks for R2:



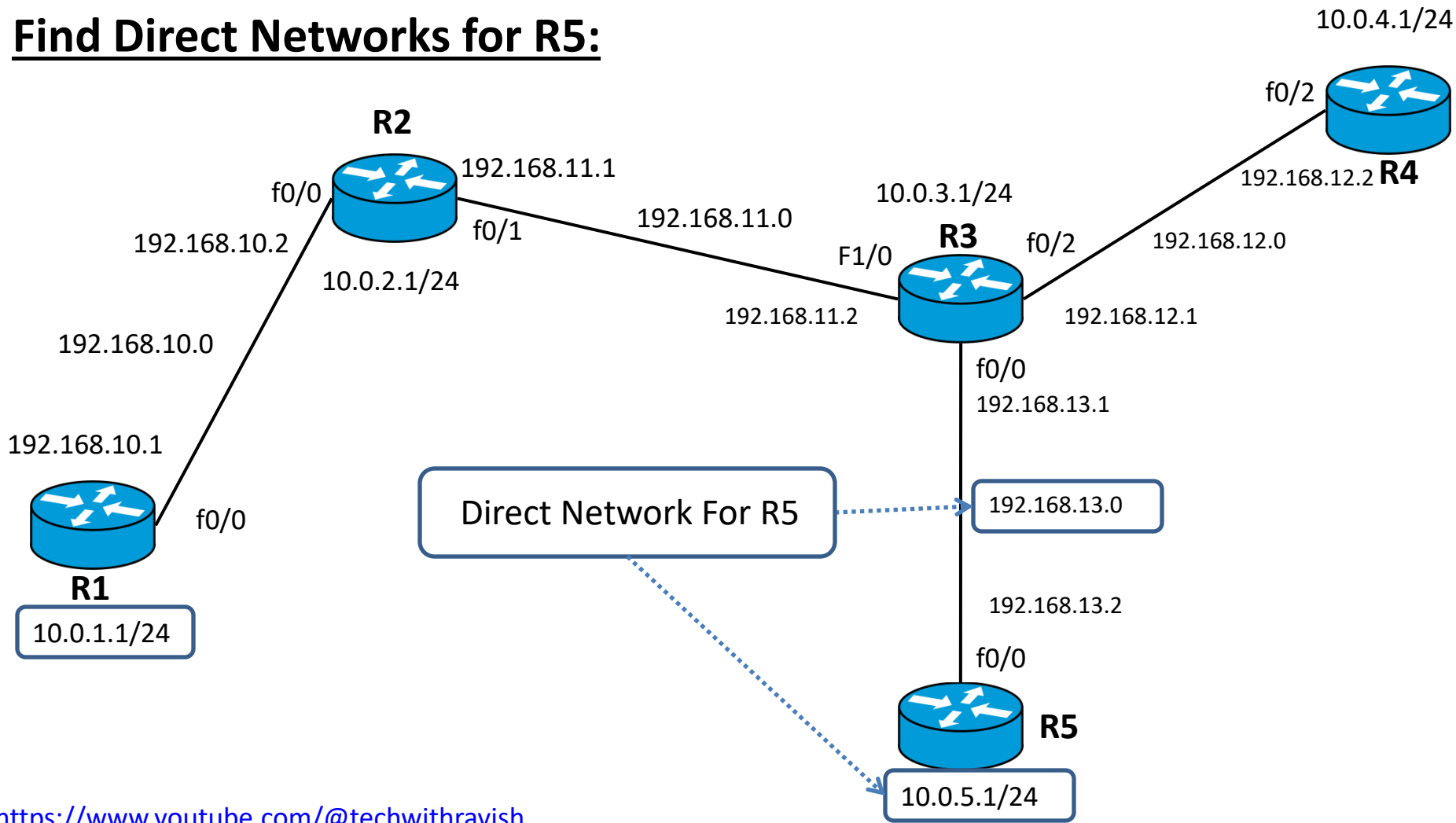
Find Direct Networks for R3:



Find Direct Networks for R4:



Find Direct Networks for R5:



Configure IP on Router 1:

R1#conf t

R1(config)#int loopback 1

R1(config-if)#ip address 10.0.1.1 255.255.255.0

R1(config-if)#Exit

R1(config)#int f0/0

R1(config-if)#no shutdown

R1(config-if)#ip address 192.168.10.1 255.255.255.0

R1(config-if)#exit

Configure IP on Router 2:

R2#conf t

R2(config)#int loopback 1

R2(config-if)#ip address 10.0.2.1 255.255.255.0

R2(config-if)#exit

R2(config)#int f0/0

R2(config-if)#no shutdown

R2(config-if)#ip address 192.168.10.2 255.255.255.0

R2(config-if)#ex

R2(config)#int f1/0

R2(config-if)#no shutdown

R2(config-if)#ip address 192.168.11.1 255.255.255.0

R2(config-if)#exit

Configure IP on Router 3:

R3#

R3#conf t

R3(config)#int loopback 1

R3(config-if)#ip address 10.0.3.1 255.255.255.0

R3(config-if)#exit

R3(config)#int f2/0

R3(config-if)#no shutdown

R3(config-if)#ip address 192.168.12.1 255.255.255.0

R3(config-if)#exit

R3(config)#int f0/0

R3(config-if)#no shutdown

R3(config-if)#ip address 192.168.13.1 255.255.255.0

R3(config-if)#exit

R3(config)#int f1/0

R3(config-if)#no shutdown

R3(config-if)#ip address 192.168.11.2 255.255.255.0

R3(config-if)#exit

Configure RIP on Router 1:

```
R4(config)#int f0/0
```

```
R4(config-if)#no shutdown
```

```
R4(config-if)#ip address 10.0.4.1 255.255.255.0
```

```
R4(config-if)#exit
```

```
R4(config)#int f2/0
```

```
R4(config-if)#no shutdown
```

```
R4(config-if)#ip address 192.168.12.2 255.255.255.0
```

```
R4(config-if)#exit
```

Configure IP on Router 5:

R5#conf t

R5(config)#int f0/0

R5(config-if)#no shutdown

R5(config-if)#ip address 192.168.13.2 255.255.255.0

R5(config-if)#exit

R5(config)#int loopback 1

R5(config-if)#ip address 10.0.5.1 255.255.255.0

R5(config-if)#exit

Configure Static Routing on Router 1 and Router 2:

R1#conf t

R1(config)#router rip

R1(config-router)#version 2

R1(config-router)#network 192.168.10.0

R1(config-router)#network 10.0.1.0

R2#conf t

R2(config)#router rip

R2(config-router)#version 2

R2(config-router)#network 192.168.10.0

R2(config-router)#network 192.168.11.0

R2(config-router)#network 10.0.2.0

Configure Static Routing on Router 1 and Router 2:

R1#conf t

R1(config)#router rip

R1(config-router)#version 2

R1(config-router)#network 192.168.10.0

R1(config-router)#network 10.0.1.0

R2#conf t

R2(config)#router rip

R2(config-router)#version 2

R2(config-router)#network 192.168.10.0

R2(config-router)#network 192.168.11.0

R2(config-router)#network 10.0.2.0

Configure Static Routing on Router 4 and Router 5:

```
R4#conf t
```

```
R4(config)#router rip
```

```
R4(config-router)#version 2
```

```
R4(config-router)#network 192.168.12.0
```

```
R4(config-router)#network 10.0.4.0
```

```
R4#conf t
```

```
R4(config)#router rip
```

```
R4(config-router)#version 2
```

```
R4(config-router)#network 192.168.13.0
```

```
R4(config-router)#network 10.0.5.0
```